

Instructions: Answer question One and any other two questions

Question One [30 marks]

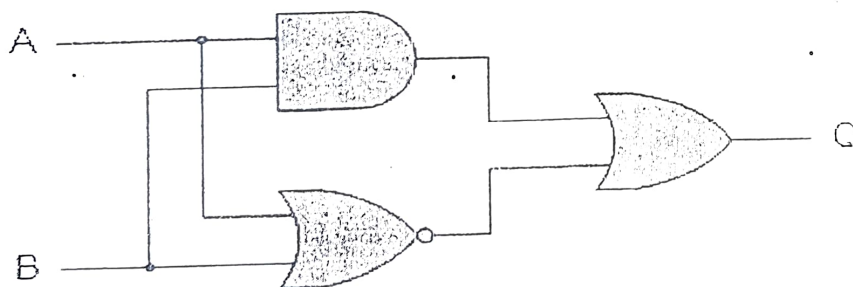
(a) Determine whether or not the vectors $A=(3,4,0)$ and $B=(-4,3,2)$ are orthogonal. [3 marks]

(b) Find the eigen values and eigen vectors associated with the matrix $\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$. [5 marks]

(c) Solve $2x+y-z=3$
 $x+y+z=1$
 $x-y-3z=4$ using Cramer's rule. [5 marks]

(d) Find the parametric and symmetric equations of the line passing through the point $(-1,2,3)$ and parallel to the vector $(3,0,-1)$ [4 marks]

(e) Find the Boolean algebra expression for the following system. [4 marks]



(f) Find the volume of the parallelepiped spanned by the vectors $a=(-2,3,1)$, $b=(0,4,0)$, and $c=(-1,3,3)$. [5 marks]

(g) Consider the matrix $A = PDP^{-1}$ where $P = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$. Compute A^5 [4 marks]

Question Two [20 marks]

(a) Solve the equations for x, y and z using inverse matrix method.

$$\begin{aligned} 5x+15y+56z &= 35 \\ -4x-11y-41z &= -26 \\ -x-3y-11z &= -7 \end{aligned}$$

[7 marks]

(b) Use Gaussian elimination method to solve for x, y and z in the system

$$\begin{aligned} x+4y-z &= -5 \\ x+y-6z &= -12 \\ 3x-y-z &= 4 \end{aligned}$$

[6 marks]

(c) Find the equation of the plane passing through the points P (1,-1,3), Q(4,1,-2), and R(-1,-1,1)

[7 marks]

Question Three [20 marks]

(a) Reduce the matrix $\begin{bmatrix} 2 & -2 & 4 & -2 \\ 2 & 1 & 10 & 7 \\ -4 & 4 & -8 & 4 \\ 4 & -1 & 14 & 6 \end{bmatrix}$ to echelon form

[7 marks]

(b) Let $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$
 $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$
 $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$.

Show that $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$

[7 marks]

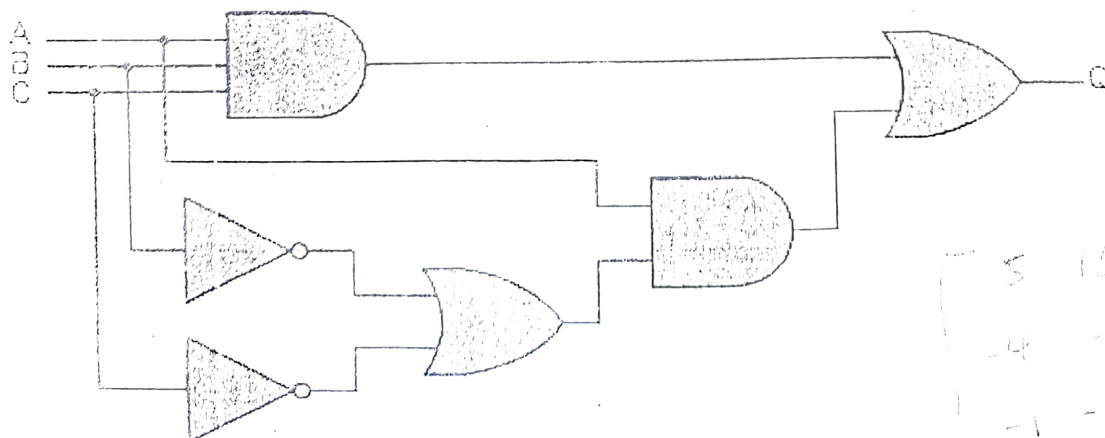
(c) Find the area of the triangle with vertices $P_1(2,2,4)$ and $P_2(-1,0,5)$ and $P_3(3,4,3)$.

[6 marks]

Question Four [20 marks]

(a) Find the Boolean algebra expression for the following system.

[6marks]



(b) Draw the logic circuit which corresponds to the Boolean expression $Y = ABC + AB' + A'B$.

[6 marks]

(c) Find the equation of the plane that passes through the point $P(1,4,3)$ and contains the line

$$\frac{x}{2} = \frac{y-1}{1} = \frac{z+1}{1}$$

(d) Let $A = \begin{pmatrix} 2 & 1 & -3 \\ -1 & 0 & 5 \end{pmatrix}$. Find AA'

[5 marks]

[3 marks]

Question Five [20 marks]

(a) Diagonalize the matrix $\begin{bmatrix} 3 & 0 & 0 \\ -3 & 4 & 9 \\ 0 & 0 & 3 \end{bmatrix}$

[8 marks]

(b) Let $A = \begin{bmatrix} 1 & 3 & 2 \\ -3 & -1 & -3 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -5 & 0 & -1 \\ 1 & 2 & -1 \\ -3 & 4 & 1 \end{bmatrix}$. Verify that $(\det A)(\det B) = \det(AB)$

[5 marks]

(c) Use properties of determinants to prove that

$$\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} = 2(a+b+c)^3$$

[7 marks]

ELECTRICAL ENGINEERING.

ECU 106: ENGINEERING MATHEMATICS III

DATE: Thursday 23rd August 2018

TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS: Answer QUESTION 1 and any OTHER TWO questions.

QUESTION 1 (Compulsory) - (30 marks)

a) Let $\underline{u} = 1, -1, 2$, $\underline{v} = (5, -2, 1)$. [2 marks]

Find $\underline{u} \cdot \underline{v}$ [2 marks]

b) Let $\underline{u} = (1, 0, -3)$, $\underline{v} = (3, -2, 1)$. [2 marks]

Find $\underline{u} \times \underline{v}$ [2 marks]

c) Find the equation of the plane passing through the points $(2, -1, 2)$, $(1, -1, 2)$, $(1, 2, 3)$. [5 marks]

d) Solve the system using Cramer's rule (method of determinants).

$$2x + 3y - z = 1$$

$$3x + 5y + 2z = 8$$

$$x - 2y - 3z = -1$$

[3 marks]

e) Let $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$. Find all eigenvalues of A and the corresponding eigenvectors

[4 marks]

f) Find the inverse of the matrix

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \quad [5 \text{ marks}]$$

g) Show that $\sim(p \vee q) \vee [\sim p \wedge q] \vee p$ is a tautology. [4 marks]

h) Use Boolean algebra to simplify $\sim(p \Rightarrow q) \vee p$ [4 marks]

QUESTION 2 (20 marks)

a) Find the volume of the parallelepiped determined by the vectors

$$\underline{u} = \underline{i} - 2\underline{j} + 3\underline{k}, \quad \underline{v} = -4\underline{i} + 7\underline{j} + 11\underline{k}, \quad \underline{w} = 5\underline{i} - 9\underline{j} - \underline{k} \quad [4 \text{ marks}]$$

b) Find the angle between the planes $x + y + z = 4$ and $x - y + z = 1$. [4 marks]

c) Solve using inverse method

$$x + 3y + 2z = 3$$

$$2x + 4y + 2z = 8$$

$$x + 2y - z = 10 \quad [4 \text{ marks}]$$

d) Two forces F_1 and F_2 act on the same body . It is known that F_1 has magnitude 3 Newtons and acts in the direction of $-\underline{i}$, whereas F_2 has magnitude 2 newtons and act in the direction of the unit vector in

$$\underline{u} = \frac{3}{5}\underline{i} - \frac{4}{5}\underline{j} \quad [4 \text{ marks}]$$

What additional force F_3 must be applied to keep the body at rest?

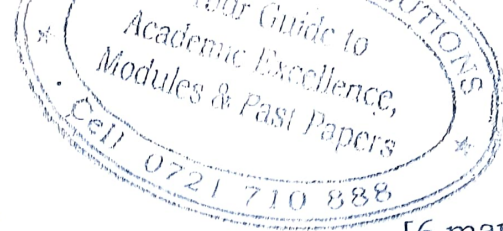
e) Let $A = \begin{pmatrix} 1 & 4 & 5 \\ 7 & -1 & 2 \\ 6 & 1 & 2 \end{pmatrix}$,

Find $\det A$. [4 marks]

QUESTION 3 (20 MARKS)

a) Diagonalize

$$A = \begin{pmatrix} 2 & 2 \\ 1 & 3 \end{pmatrix} \quad [6 \text{ marks}]$$



b) Use Gaussian elimination to solve

$$2x + y - 2z = 10$$

$$3x + 2y + 2z = 1$$

$$5x + 4y + 3z = 4$$

[6 marks]

c) Given $A = \begin{pmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{pmatrix}$

i) Reduce A to echelon form

[3 marks]

ii) Reduce A to row canonical form

[3 marks]

d) Let $A = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ 3 & 2 \end{pmatrix}$ and $C = \begin{pmatrix} 2 & -1 \\ 4 & 3 \end{pmatrix}$

Find $AB^t + 2C$.

[2 marks]

QUESTION 4 (20 marks)

a) Find the parametric and the symmetric equations of a line passing through the points $(5, 4, -3)$ and $(1, -3, 2)$.

[5 marks]

b) Determine whether the following lines intersect, are parallel or skew

$$L_1: \frac{x-1}{2} = \frac{y+1}{1} = \frac{z-2}{1} \text{ and}$$

$$L_2: \frac{x+2}{4} = \frac{y}{-3} = \frac{z-1/2}{-1}$$

[5 marks]

c) Find the equation of the line passing through $(-1, 2, 3)$ that is parallel to the line of intersection of the planes $3x - 2y + z = 4$ and $x + 2y + 3z = 5$.

[5 marks]

d) Find the distance from the point $(3, -8, 1)$ to the line.

$$\frac{x-3}{3} = \frac{y+7}{-1} = \frac{z+2}{5}$$

[5 marks]

QUESTION 5 (20 marks)

- a) Use Boolean algebra to simplify

$$\sim(p \vee q) \vee (\sim p \wedge q)$$

[5 marks]

- b) Find the converse of the contrapositive of $p \Rightarrow \sim p$

[5 marks]

- c) Use the truth table to show whether the following proposition is a tautology, contradiction or logically indeterminate.

$$(p \wedge q) \Rightarrow (p \vee q)$$

[5 marks]

- d) For the equation $z = xy + \bar{w}y$ construct a logic gate.

[5 marks]

ECU 106: ENGINEERING MATHEMATICS III

DATE: Wednesday, 14th June 2017

TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS:

Answer Question 1 and any other two questions.

QUESTION 1 - COMPULSORY (30 MARKS)

- (a) $\underline{u} = (3, -2, 1)$ and $\underline{v} = (7, 1, -3)$. Find $\underline{u} \cdot \underline{v}$ (2 marks)
- (b) Let $\underline{a} = (1, -2, 4)$ and $\underline{b} = (2, -1, -2)$. Find $\underline{a} \times \underline{b}$ (3 marks)
- (c) Find the equation of the plane passing through points $(-1, 2, 1)$, $(0, -3, 2)$ and $(1, 1, -4)$. $n \cdot (r - r_0) = 0$ (5 marks)
- (d) Given that $A = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}$ determine C such that
 $C = (A + B)^2 - (A^2 + 2AB + B^2)$ (3 marks)
- (e) Use the Cramer's rule (method of determinants) to solve the system
 $2x + 3y - z = 1$
 $3x + 5y + 2z = 8$
 $x - 2y - 3z = -1$
- (f) Use Boolean algebra to simplify $\sim(p \vee \sim q) \wedge q$ (4 marks)
- (g) Use truth tables to show that $(p \wedge q) \Rightarrow (p \vee q)$ is a tautology. (4 marks)
- (h) Find the eigenvalues and eigenvectors of the matrix $A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$ (4 marks)

QUESTION 2 (20 MARKS)

- (a) (i) Determine k so that the vectors $\underline{u} = (1, k, -3)$ and $\underline{v} = (2, -5, 4)$ are orthogonal. (3 marks)
- (ii) If $\underline{u} = (2, -7, 1)$, $\underline{v} = (-3, 0, 4)$, $\underline{w} = (0, 5, -8)$, find $2\underline{u} + 3\underline{v} - 5\underline{w}$ (3 marks)
- b) Find the parametric and the symmetric equations of the line which passes through $(4, -1, 9)$ and perpendicular to the plane $3x - 2y + z = 18$. (5 marks)
- (c) Find the angle between the vectors $\underline{v} = \underline{i} + \underline{j} + \underline{k}$ and $\underline{w} = \underline{i} - \underline{j} + \underline{k}$. (3 marks)
- (d) Find the distance from the point $(3, -8, 1)$ to the line $\frac{x-3}{3} = \frac{y+7}{-1} = \frac{z+2}{5}$. (6 marks)

QUESTION 3 (20 MARKS)

- (a) (i) Find the determinant of the matrix $A = \begin{pmatrix} 2 & -4 & 1 \\ 1 & -2 & 3 \\ 5 & 1 & -1 \end{pmatrix}$ (2 marks)
- (ii) Find the inverse of the matrix $A = \begin{pmatrix} 2 & 3 & -4 \\ 0 & -4 & 2 \\ 1 & -1 & 5 \end{pmatrix}$ (5 marks)
- (b) Solve
- $$\begin{aligned} x + y + z &= 5 \\ x + 2y + 3z &= 10 \\ 2x + y + z &= 6 \end{aligned}$$
- using inverse matrix method. (8 marks)
- (c) Solve
- $$\begin{aligned} 2x + y - 2z &= 10 \\ 3x + 2y + 3z &= 10 \\ 5x + 4y + 3z &= 4 \end{aligned}$$
- (5 marks)

Using Gauss elimination method.

QUESTION 4 (20 MARKS)

- (a) Find the equation of the plane which passes through the point $(1, 2, -1)$ and parallel to the plane $2x - 3y + z = 1$. (3 marks)

- (b) Find all eigenvalues and eigenvectors of the matrix $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 4 \end{pmatrix}$ (9 marks)

- (c) Diagonalize the matrix $A = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{pmatrix}$. (8 marks)

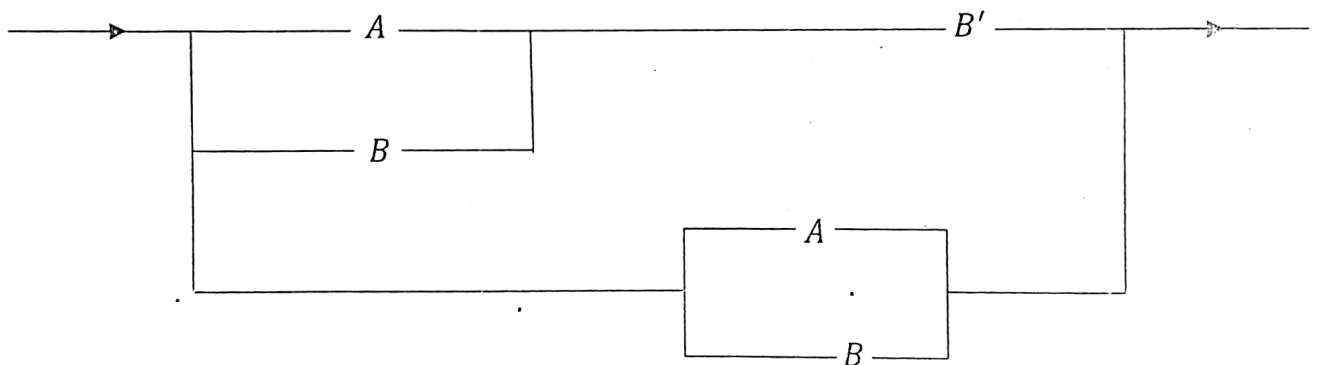
QUESTION 5 (20 MARKS)

- (a) Use Boolean algebra to simplify $(p \wedge q) \vee (p \wedge \sim q) \vee (\sim p \wedge q)$ (3 marks)
- (b) By using truth tables, show whether the following propositions are logically true, logically false or neither

(i) $(p \vee q) \wedge \sim p$ (3 marks)

(ii) $\sim (p \vee q) \Rightarrow \sim p$ (4 marks)

- (c) Given the circuit



- (i) Write down the Boolean function of the circuit. (2 marks)
- (ii) Simplify the Boolean function in (i) and draw the corresponding simplified circuit. (6 marks)

- (d) For the equation $z = xy + w'y$, construct a logic gate. (2 marks)

Answer Question 1 and any other two questions

1. a) i. Determine if the following vector is parallel, orthogonal or neither $\vec{u} = 2\vec{i} - \vec{j}$ and

$$\vec{v} = -\frac{1}{2}\vec{i} + \frac{1}{4}\vec{j} \quad (2 \text{ marks})$$

- ii. Compute the dot product of the following; $\vec{v} = 5\vec{i} - 8\vec{j}$, $\vec{w} = \vec{i} + 2\vec{j}$ (1 mark)

- iii. Determine the angle between; $\vec{a} = \langle 2, 0, 3 \rangle$, $\vec{b} = \langle 3, 2, 4 \rangle$ (2 marks)

- iv. Find the work done by the force $F = \langle 3, 1, 5 \rangle$ in moving a particle from P to Q where position vectors of P and Q are $\langle 1, 2, 3 \rangle$, and $\langle 2, 4, 4 \rangle$ respectively. (2 marks)

- v. Determine if the following vectors are coplanar; $3\vec{i} + 2\vec{j} - 2\vec{k}$,

$$5\vec{i} - 3\vec{j} + 3\vec{k}, \quad 5\vec{i} - \vec{j} + \vec{k} \quad (3 \text{ marks})$$

- (b) Determine the line that passes through the point (0, -3, 8) and is parallel to the line given by $x = 10 + 3t$, $y = 12t$, and $z = -3 - t$ and passes through the xy plane and if it does give the coordinates of that point. (4 marks)

- (c) Find the equation of the line r_1 through (0, 1, -2) and (3, 4, 3) and the equation of the line r_2 through the points (1, 1, 0), (-3, -2, -7). Determine if r_1 and r_2 intersect (5 marks)

- (d) Find the equation of the plane through the points (0, 1, 1), (1, 0, 1) and (1, 1, 0) (3 marks)

- (e) Find the volume of the parallelepiped whose three edges meeting at a point are; $2\vec{i} - 3\vec{j} + 4\vec{k}$, $\vec{i} + 2\vec{j} - \vec{k}$ and $3\vec{i} - \vec{j} + 2\vec{k}$ (3 marks)

- (f) If $\vec{a} = \vec{i} - 2\vec{j} + 3\vec{k}$, $\vec{b} = 2\vec{i} + 3\vec{j}$, $\vec{c} = 3\vec{i} - \vec{k}$; Find (5 marks)

i. $\vec{a} \cdot (\vec{b} \times \vec{c})$

ii. $\vec{a} \times (\vec{b} \times \vec{c})$

2. (a) Solve the following set of linear equations by the matrix determinant method;

$$x + y + z - 4 = 0$$

$$2x - 3y + 4z - 33 = 0$$

$$3x - 2y - 2z - 2 = 0$$

(10 marks)

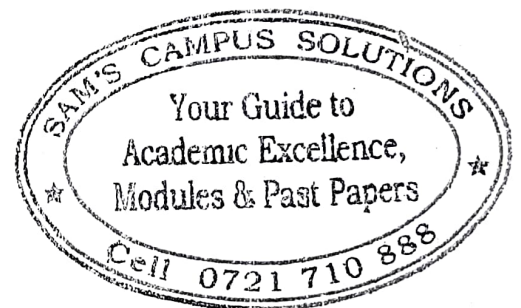
- b) The relationship between the displacement, s , velocity, v , and acceleration, a , of a piston is given by equations:

$$s + 2v + 2a = 4$$

$$3s - v + 4a = 25$$

$$3s + 2v - a = -4$$

Use matrices to determine the values of s , v , and a



3. (a) Using the Gaussian elimination method solve the following system of linear equations;

$$x = -y + 5$$

$$-2x + 2z = y - 10$$

$$3x + 6y + 7z = 14$$

(6 marks)

- (b) Solve the following system of linear equations using the Cramer's rule

$$x + 2y + 3z = 5$$

$$2x - 3y - z = 3$$

$$-3x + 4y + 5z = 3$$

(8 marks)

- (c) Determine the inverse of the matrix;

$$\begin{bmatrix} 3 & 4 & -1 \\ 2 & 0 & 7 \\ 1 & -3 & -2 \end{bmatrix}$$

(6 marks)

4. (a) Find the Eigen value and the Eigen vectors of the following matrix ;

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

(10 marks)

- (b) Diagonalize the following matrix if possible;

$$\begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

(10 marks)

5. (a) Use the logic gates to represent; the OR, AND and NOT function of the Boolean algebra;

draw also their truth tables

(8 marks)

- (b) Construct the switching circuit to meet the requirements of the Boolean expression

$$z = A \cdot \bar{C} + \bar{A} \cdot B + \bar{A} \cdot B \cdot \bar{C} \quad (12 \text{ marks})$$